(1) $\sim(\sim P \wedge \sim q)$

Sol:

P	q	$\sim P$	$\sim q$	$(\sim P \wedge \sim q)$	$\sim(\sim P \wedge \sim q)$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	F

(2) $q \leftrightarrow (\sim P \vee \sim q)$

Sol:

P	q	$\sim P$	$\sim q$	$(\sim P \vee \sim q)$	$q \leftrightarrow (\sim P \vee \sim q)$
T	F	F	F	F	F
T	F	F	T	T	F
F	T	T	F	T	T
F	F	T	T	T	F

3) $(P \to q) \wedge (\sim P \to q)$

Sol:

P	q	$\sim P$	$(\sim P \to q)$	$P \to q$	$(P \to q) \wedge (\sim P \to q)$
T	T	F	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	F	T	F	T	F

4) $\sim(P \to q) \leftrightarrow (\sim P \to \sim q)$

Sol:

P	q	$\sim P$	$\sim q$	$(\sim P \to \sim q)$	$P \to q$	$\sim(P \to q)$	$\sim(P \to q) \leftrightarrow (\sim P \to \sim$
T	T	F	F	T	T	F	F
T	F	F	T	T	F	T	T
F	T	T	F	F	T	F	T
F	F	T	T	T	T	F	F

5) $(\sim p \wedge (\sim q \wedge p)) \vee (q \wedge r) \vee (p \wedge r)$

Soln

p	q	r	$\sim p$	$\sim q$	$q \wedge r$	$p \wedge r$	$(q \wedge r) \vee (p \wedge r)$
T	T	T	F	F	T	T	T
T	T	F	F	F	F	F	F
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	F
F	T	T	T	F	T	F	T
F	T	F	T	F	F	F	F
F	F	T	T	T	F	F	F
F	F	F	T	T	F	F	F

$\sim q \wedge p$	$(\sim p \wedge (\sim q \wedge p))$	$(\sim p \wedge (\sim q \wedge p)) \vee (q \wedge r) \wedge (p \wedge r)$
F	F	T
F	F	F
T	F	T
T	F	F
F	F	T
F	F	F
F	F	F
F	F	F

6) $(p \downarrow q) \vee (p \wedge q)$

Soln

p	q	$p \wedge q$	$p \vee q$	$p \downarrow q$	$(p \downarrow q) \vee (p \wedge q)$
T	T	T	T	F	T
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	T

Verify 1) Demorgen laws.

$\sim(p \wedge q) = \sim p \vee \sim q$.

Sol:-

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \vee \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Same.

$p \rightarrow (\sim q \vee r)$.

p	q	r	$\sim q$	$(\sim q \vee r)$	$p \rightarrow (\sim q \vee r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	F
F	F	T	T	T	T
F	F	F	T	T	T

Prove that: $(p\downarrow q)\vee(p\downarrow\sim q)\vee p$ is tautology.

Sol:

p	q	$\sim q$	$p\vee q$	$p\downarrow q$	$p\vee\sim q$	$p\downarrow\sim q$
T	T	F	T	F	T	F
T	F	T	T	F	T	F
F	T	F	T	F	F	T
F	F	T	F	T	T	F

$(p\downarrow q)\vee(p\downarrow\sim q)\vee p$
T
T
T
T

It is tautology.

Prove that: $(p\wedge q)\vee r \leftrightarrow ((\sim p\vee\sim q)\wedge\sim r)$ is contradiction

p	q	r	$\sim p$	$\sim q$	$\sim r$	$\sim p\vee\sim q$	$(\sim p\vee\sim q)\wedge\sim r$
T	T	T	F	F	F	F	F
T	T	F	F	F	T	F	F
T	F	T	F	T	F	T	F
T	F	F	F	T	T	T	T
F	T	T	T	F	F	T	F
F	T	F	T	F	T	T	T
F	F	T	T	T	F	T	F
F	F	F	T	T	T	T	T

$p\wedge q$	$(p\wedge q)\vee r$	$(p\wedge q)\vee r \leftrightarrow ((\sim p\vee\sim q)\wedge\sim r)$
T	T	F
T	T	F
F	T	F
F	F	F
F	T	F
F	F	F
F	T	F
F	F	F

(Yes it is contradiction.